Vector Calculus and Complex Analysis Classwork (2025/4/27)

Name:

Student ID:

Find the values of

$$\int_{\gamma} x - y + ix^2 \, dz$$

where $z = x + iy$ and γ is:

(i) the straight line joining 0 to $1 + i$;

(ii) the imaginary axis from 0 to i;

(iii) the line parallel to the real axis from i to $1 + i$.

Solution

Let $f(z) = x - y + ix^2$ where $z = x + iy$.

(i) The straight line from 0 to $1 + i$ has parametrisation $\gamma(t) = t + it$, $0 \leq t \leq 1$. Here $\gamma'(t) = 1 + i$ and $f(\gamma(t)) = t - t + it^2 = it^2$. Hence

$$\begin{aligned}\int_\gamma x - y + ix^2 &= \int_0^1 it^2(1+i)\,dt \\ &= \int_0^1 -t^2 + it^2\,dt \\ &= \left[\frac{-1}{3}t^3 + \frac{i}{3}t^3\right]_0^1 \\ &= \frac{-1}{3} + \frac{i}{3}.\end{aligned}$$

(ii) The imaginary axis from 0 to i has parametrisation $\gamma(t) = it$, $0 \leq t \leq 1$. Here $\gamma'(t) = i$ and $f(\gamma(t)) = -t$. Hence

$$\begin{aligned}\int_\gamma x - y + ix^2 &= \int_0^1 -it\,dt \\ &= \left[\frac{-it^2}{2}\right]_0^1 \\ &= \frac{-i}{2}.\end{aligned}$$

(iii) The line parallel to the real axis from i to $1 + i$ has parametrisation $\gamma(t) = t + i$, $0 \leq t \leq 1$. Here $\gamma'(t) = 1$ and $f(\gamma(t)) = t - 1 + it^2$. Hence

$$\begin{aligned}\int_\gamma x - y + ix^2 &= \int_0^1 (t - 1 + it^2)\,dt \\ &= \left[\frac{t^2}{2} - t + \frac{it^3}{3}\right]_0^1 \\ &= \frac{-1}{2} + \frac{i}{3}.\end{aligned}$$